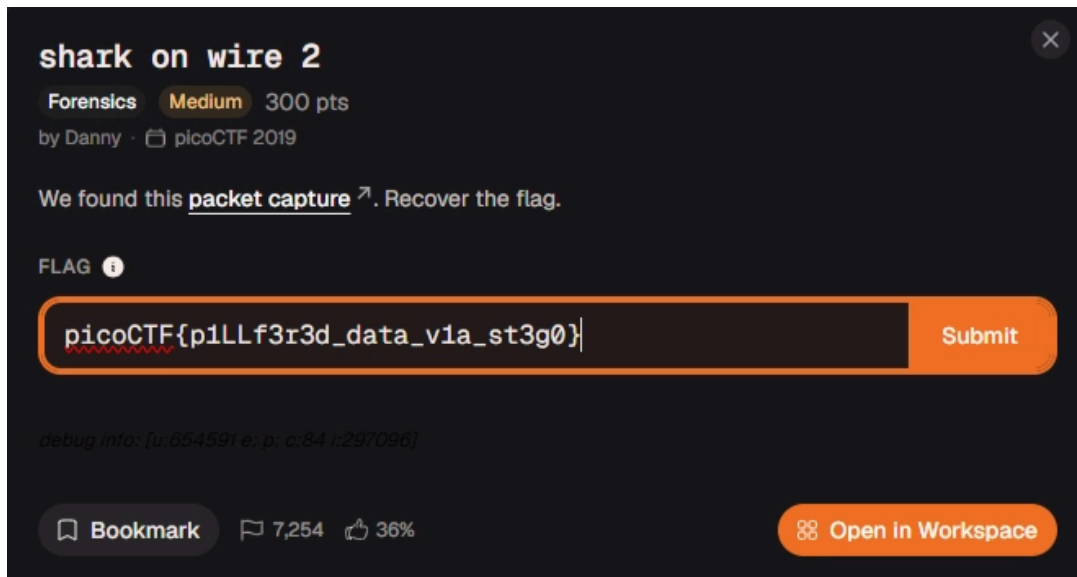


shark on wire 2 | picoCTF Writeup

Written by: Alexander Sapo



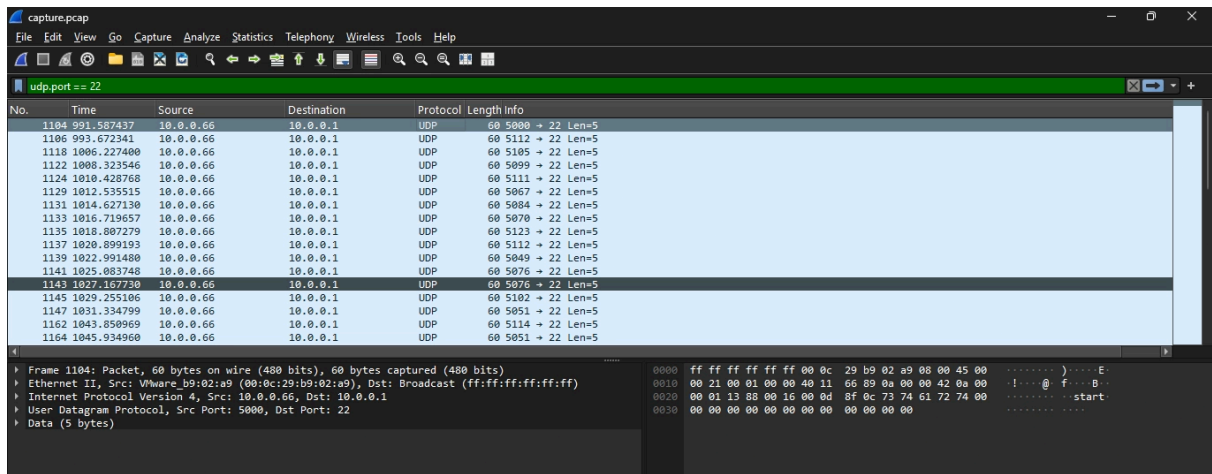
Process:

During my initial analysis, the traffic appeared normal until I noticed that one IP address was communicating with different ports on the same IP address, which I found unusual.

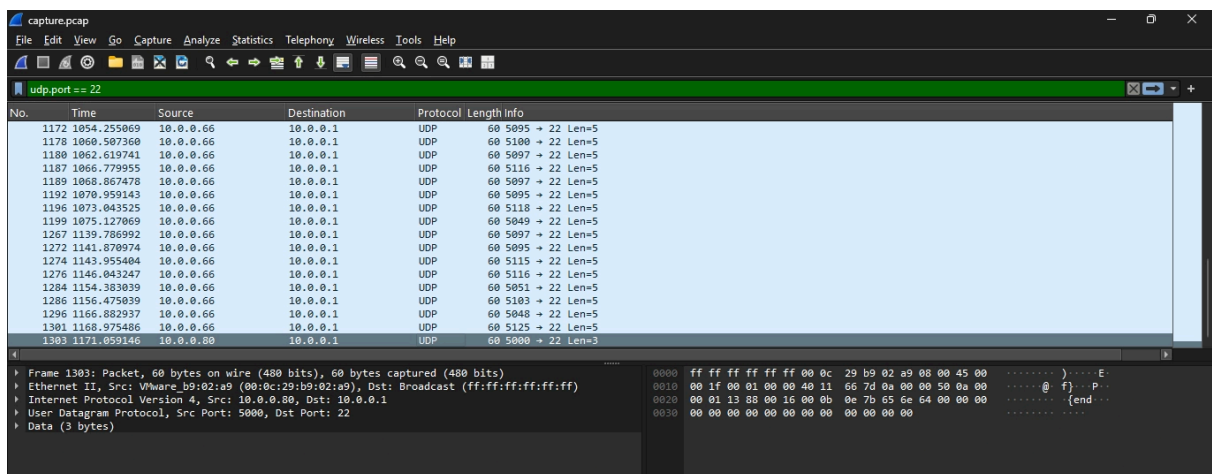
While analyzing the network traffic, we noticed packets using UDP destination port 22. As shown in the packet details, the first relevant packet contains the text "**start**", while the last packet contains "**end**". This served as a strong hint that the packets between these markers contained the hidden data. We then filtered the capture using `udp.dstport == 22` to focus on the relevant traffic.

```
udp.port == 22
```

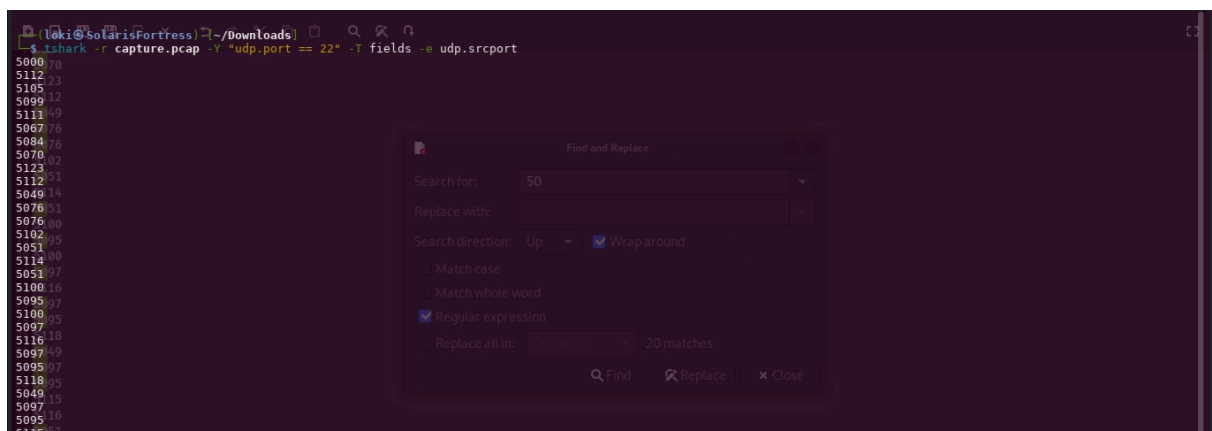
```
start
```



end

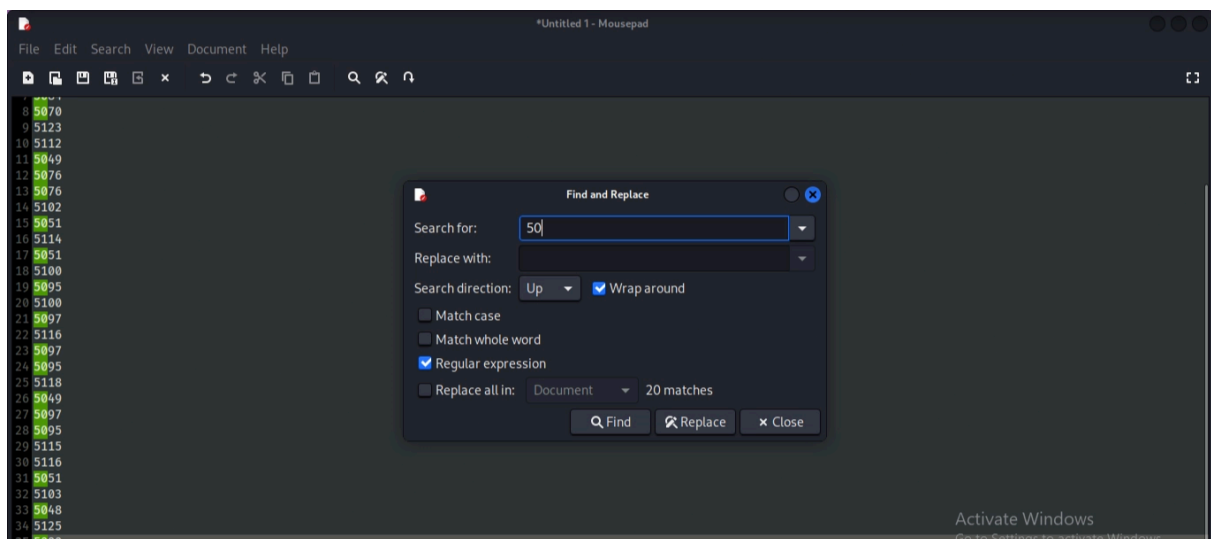


Filtered the traffic in TShark to display only packets with destination UDP port 22 and their corresponding source ports.



As I reviewed the ports, I guessed that they could have been used as a covert communication channel or to transfer data without being easily noticed. However, I did not realize the intended technique during my analysis, so I read the writeup.

The writeup explained that the port numbers represented character codes, except for the extra trailing digit, which was either **50** or **5**. I removed the trailing digit from each port number, then pasted the remaining numbers into CyberChef as space-separated values. Using the **From Charcode** operation with **Base 10**, I successfully decoded the data and recovered the flag.



This was a valuable learning experience, as it showed that an attacker can hide data within seemingly legitimate network metadata, such as port numbers, to create a covert communication channel.

